MILITARY SPECIFICATION

MOTOR-DETONATOR ASSEMBLY, ROCKET, INERT, FOR DEMOLITION KIT, CRATERING, TRAINING, XM270, LOADING, ASSEMBLING AND PACKING

This specification is approved for use by US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of Motor-detonator assembly designated as Motor-detonator Assembly, Rocket, Inert for Demolition Kit, Cratering, Training, XM270.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1188- Commercial Packaging of Supplies and Equipment

FSC 1375

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.5)

U.S. ARMY RESEARCH AND DEVELOPMENT COMMAND

9331232 - Box, Packing, Wood, Nailed
9331651 - Rocket Motor Detonator Assembly (Inert)
9332620 - Packing and Marking for Box, Packing, Wood Nailed, for Demolition Kit, Cratering, Training, XM270 (Inert)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Components, sub-assemblies and assemblies. The components, sub-assemblies and assemblies shall comply with all requirements specified on Dwg. 9331651 and associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. All parts, sub-assemblies and assemblies shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, scales, corrosion and other foreign matter. The cleaning method used shall not be detrimental to any of the parts nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined. Surface coatings shall be continuous except for a few light scratches which do not expose base material. Paintings and coatings on all components, sub-assemblies and assemblies shall be restored when such paintings and coatings have been marred or obliterated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated below:

Part Description	Drawing	Quantity
Rocket Motor Detonator Assembly (Inert)	9331651	10
Detonator Assembly (Inert)	9331653	10
Rocket Motor Assembly (Inert)	9331652	10
Clamp Conduit	9244410	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First Article Inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63375(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Rocket Motor Detonator Assembly (Inert)				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Rocket Motor Assembly (Dwg. 9331652) Examination for defects	10	-	3.2	4.4.2.9
	Detonator Assembly (Dwg. 9331653) Examination for defects	10	-	3.2	4.4.2.4
	Rocket Motor Detonator Assembly (Dwg. 9331651) Examination for defects	10	-	3.2	4.4.2.11
	Clamp Conduit (Dwg. 9244410) Examination for defects	10	-	3.2	4.4.2.10

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Rocket Motor Assemblies shall contain:

a. Parts and assemblies from not more than one interfix lot number from one manufacturer.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and test tables, sampling plans with major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63375(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Body Protector, Rocket	1 OF 1	9244426
			NEXT HIGHER ASSEMBLY 9331653

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter max.		0.40%	3.2	Gage
102	Outside diameter of flange max.		0.40%	3.2	Gage
103	Thickness of flange max.		0.40%	3.2	Gage
104	Snap ring groove missing		0.40%	3.2	Visual
105	Minor diameter min.		0.40%	3.2	Gage
106	Pitch diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63375(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Flange, Rocket	1 OF 1	9244447
			NEXT HIGHER ASSEMBLY
			9331653

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter min.		0.40%	3.2	Gage
102	Diameter of smaller countersink min.		0.40%	3.2	Gage
103	Depth of smaller countersink min.		0.40%	3.2	Gage
104	Groove missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63375(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.3	Retainer, Protector Rocket	1	1	9244427
				NEXT HIGHER ASSEMBLY: 9331653

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter min.		0.40%	3.2	Gage
102	Pitch diameter		0.40%	3.2	Gage
103	Major diameter max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63375(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Detonator Assembly (Inert)	1 OF 1	9331653
			NEXT HIGHER ASSEMBLY 9331651

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any part missing or incorrectly assembled		0.40%	3.2	Visual
102	Marking missing or incorrect		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63375(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Plate, Retainer, Rocket	1 OF 1	9244408
			NEXT HIGHER ASSEMBLY 9331652

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63375(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Head, Rocket	1 OF 1	9244407
			NEXT HIGHER ASSEMBLY 9331652

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter max.		0.40%	3.2	Gage
102	Inside diameter min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63375(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Plug, Head	1 OF 1	9285387
			NEXT HIGHER ASSEMBLY 9331652

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter		0.40%	3.2	Gage
102	Minor diameter min.		0.40%	3.2	Gage
103	Outside diameter max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63375(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.8	Rocket Motor Case Assembly	1 OF 1	9244441
			NEXT HIGHER ASSEMBLY 9331652

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of smallest countersink min.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63375(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Rocket Motor Assembly (Inert)	1 OF 1	9331652
			NEXT HIGHER ASSEMBLY: 9331651

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any part missing or incorrectly assembled		0.40%	3.2	Visual/Manual
102	Marking missing or inaccurate		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.10	Clamp, Conduit	1 OF 1	9244410
			NEXT HIGHER ASSEMBLY
			9331651

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Radius of clamping surface		0.40%	3.2	Gage
102	Location of holes		0.40%	3.2	Gage
103	Diameter of holes		0.40%	3.2	Gage
Minor					
201	Angle		0.65%	3.2	Gage
202	Overall length		0.65%	3.2	Gage
203	Protective finish missing or damaged		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Rocket Motor Detonator Assembly (Inert)	1 OF 1	9331651
			NEXT HIGHER ASSEMBLY: 9331717

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect trim at wire ends prior to crimping parallel splices		0.40%	3.2	Gage
102	Any part missing or not assembled properly		0.40%	3.2	Visual/Manual
Minor					
201	Adhesive tape missing or not applied properly		0.65%	3.2	Visual/Manual
202	Marking missing, incorrect or illegible		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Packing Box, prior to securing top	1 OF 1	9332620
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHO
Critical	None defined				
Major					
101	Descriptive nomenclature incorrect		0.40%	3.2	Visual
102	Contents positioned improperly		0.40%	3.2	Visual/Manual
103	Insufficient area for subsequent seals		0.40%	3.2	Visual
104	End, side or top fillers missing		0.40%	3.2	Visual
Minor					
201	Foreign material in box		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.5	Visual

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test method and procedures

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained cy current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. The visual, manual and mechanical inspections and tests shall verify compliance with requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

5. PACKAGING

5.1 Preservation and packaging. Packaging shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be in accordance with drawing 9332620.

5.3 Marking. Marking shall be in accordance with drawing 9332620.

6. NOTES

6.1 Intended use. The booster assembly covered by this specification is used in the Rocket Motor Assembly and is a component of the Demolition Kit Cratering, Training, XM270.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, U.S. Army Research and Development Command, ATTN: DRDAR-QAA-I, Dover, New Jersey 07801.

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6.4 Distribution of ammunition data cards. Distribution of data cards shall include the following: Commander, ARRADCOM, ATTN: DRDAR-QAA-M, Dover, New Jersey 07801.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the compliance of ARRADCOM.

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Preparing Activity:
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